

COMP9319 Web Data Compression and Search

BWT, MTF and Pattern Matching

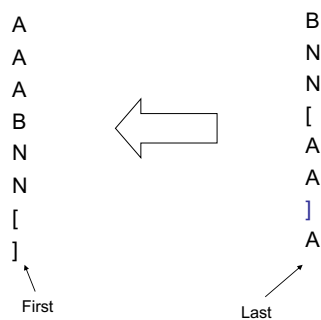
1

BWT

- Burrows–Wheeler transform (BWT) is an algorithm used to prepare data for use with data compression techniques such as bzip2.
- It was invented by Michael Burrows and David Wheeler in 1994 at DEC SRC, Palo Alto, California.
- It is based on a previously unpublished transformation discovered by Wheeler in 1983.

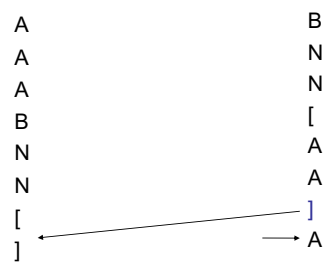
2

Recall: Last column = BWT



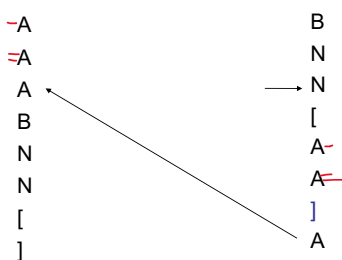
2

A]



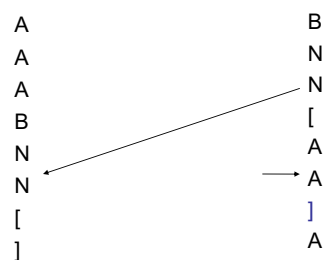
3

NA]



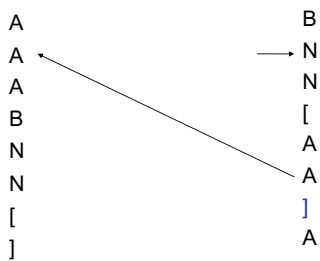
4

ANA]



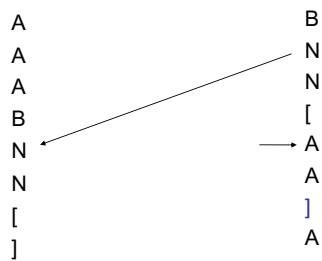
5

NANA]



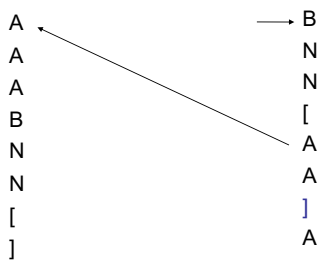
6

ANANA]



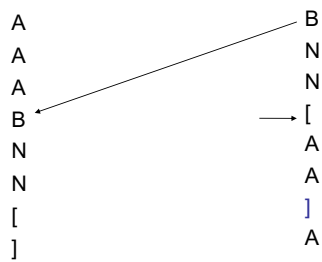
7

BANANA]



8

[BANANA]



9

Example using C[] & Occ[]

Position	Symbol	# Matching
0	B	0
1	N	0
2	N	1
3	[	0
4	A	0
5	A	1
6	]	0
7	A	2

Symbol	# LessThan
A	0
B	3
N	4
[	6
]	7

10

???????]

Position	Symbol	# Matching
0	B	0
1	N	0
2	N	1
3	[	0
4	A	0
5	A	1
6	]	0
7	A	2

Symbol	# LessThan
A	0
B	3
N	4
[	6
]	7

11

?????A]

Position	Symbol	# Matching
0	B	0
1	N	0
2	N	1
3	[	0
4	A	0
5	A	1
6	]	0
7	A	2

Symbol	# LessThan
A	0
B	3
N	4
[	6
]	7

12

?????NA]

Position	Symbol	# Matching
0	B	0
1	N	0
2	N	1
3	[	0
4	A	0
5	A	1
6	]	0
7	A	2

Symbol	# LessThan
A	0
B	3
N	4
[	6
]	7

13

????ANA]

Position	Symbol	# Matching
0	B	0
1	N	0
2	N	1
3	[	0
4	A	0
5	A	1
6	]	0
7	A	2

Symbol	# LessThan
A	0
B	3
N	4
[	6
]	7

14

???NANA]

Position	Symbol	# Matching
0	B	0
1	N	0
2	N	1
3	[	0
4	A	0
5	A	1
6	]	0
7	A	2

Symbol	# LessThan
A	0
B	3
N	4
[	6
]	7

15

??ANANA]

Position	Symbol	# Matching
0	B	0
1	N	0
2	N	1
3	[	0
4	A	0
5	A	1
6	]	0
7	A	2

Symbol	# LessThan
A	0
B	3
N	4
[	6
]	7

16

?BANANA]

Position	Symbol	# Matching
0	B	0
1	N	0
2	N	1
3	[	0
4	A	0
5	A	1
6	]	0
7	A	2

Symbol	# LessThan
A	0
B	3
N	4
[	6
]	7

17

[BANANA]

Position	Symbol	# Matching	Symbol	# LessThan
0	B	0	A	0
1	N	0	B	3
2	N	1	N	4
3	[	0	[	6
4	A	0	]	7
5	A	1		
6	]	0		
7	A	2		

18

[BANANA]

Position	Symbol	# Matching	Symbol	# LessThan
0	B	0	A	0
1	N	0	B	3
2	N	1	N	4
3	[	0	[	6
4	A	0	]	7
5	A	1		
6	]	0		
7	A	2		

Occ / Rank

c[i]

19

Move to Front (MTF)

Reduce entropy based on local frequency correlation

Usually used for BWT before an entropy-encoding step

Author and detail:

Original paper at webcms3

http://www.arturocampos.com/ac_mtf.html

3

Example: abaabacad

Symbol	Code	List
a	0	abcde.....
b	1	bacde.....
a	1	abcde.....
a	0	abcde.....
b	1	bacde.....
a	1	abcde.....
c	2	cabde.....
a	1	acbde.....
d	3	dacbe.....

To transform a general file, the list has 256 ASCII symbols.

4

Example: abaaabbbccdddddcc

Symbols: abaaabbbccdddddcc

Codes (in ASCII binary): 01100001, 01100010, 01100001, 01100001, ..., 01100100, 01100011, 01100011

Codes (in ASCII dec): 97, 98, 97, 97, 97, 98, 98, 98, 99, 99, 100, 100, 100, 100, 99, 99

5

Example: abaaabbbccdddddcc

Symbols: abaaabbbccdddddcc

Codes (in ASCII binary): 01100001, 01100010, 01100001, 01100001, ..., 01100100, 01100011, 01100011

Codes (in ASCII dec): 97, 98, 97, 97, 97, 98, 98, 98, 99, 99, 100, 100, 100, 100, 99, 99

Recall that Shannon's entropy reaches the max when there is max uncertainty, i.e., equal probability, like the example above (4 "97"s, 4 "98"s, 4 "99"s, 4 "100"s).

e.g., Entropy $H = 2.00$

6

Example: abaaabbbccdddddcc

Symbols: abaaabbbccdddddcc

Codes (in ASCII binary): 01100001, 01100010, 01100001, 01100001, ..., 01100100, 01100011, 01100011

Codes (in ASCII dec): 97, 98, 97, 97, 97, 98, 98, 98, 99, 99, 100, 100, 100, 100, 99, 99

List

Index: 0 1 2 3 4 97 98 99 100 101 102 ... 255

Value: a b c d e f

Codes (in ASCII): 97, 98, 97, 97, 97, 98, 98, 98, 99, 99, 100, 100, 100, 100, 99, 99

Codes (in MTF): 97

7

Example: abaaabbbccdddddcc

Symbols: abaaabbbccdddddcc

Codes (in ASCII binary): 01100001, 01100010, 01100001, 01100001, ..., 01100100, 01100011, 01100011

Codes (in ASCII dec): 97, 98, 97, 97, 97, 98, 98, 98, 99, 99, 100, 100, 100, 100, 99, 99

List

Index: 0 1 2 3 4 97 98 99 100 101 102 ... 255

Value: a b c d e f

Codes (in ASCII): 97, 98, 97, 97, 97, 98, 98, 98, 99, 99, 100, 100, 100, 100, 99, 99

Codes (in MTF): 97

List

Index: 0 1 2 3 4 97 98 99 100 101 102 ... 255

Value: a b c d e f

8

Example: abaaabbbccdddddcc

List

Index: 0 1 2 3 4 97 98 99 100 101 102 ... 255

Value: a b c d e f

Codes (in ASCII): 97, 98, 97, 97, 97, 98, 98, 98, 99, 99, 100, 100, 100, 100, 99, 99

Codes (in MTF): 97, 98

List

Index: 0 1 2 3 4 97 98 99 100 101 102 ... 255

Value: b a c d e f

9

Example: abaaabbbccdddddcc

List

Index: 0 1 2 3 4 97 98 99 100 101 102 ... 255

Value: b a c d e f

Codes (in ASCII): 97, 98, 97, 97, 97, 98, 98, 98, 99, 99, 100, 100, 100, 100, 99, 99

Codes (in MTF): 97, 98, 1

List

Index: 0 1 2 3 4 97 98 99 100 101 102 ... 255

Value: b a c d e f

10

Example: abaaabbbccdddddcc

List

Index: 0 1 2 3 4 97 98 99 100 101 102 ... 255

Value: a b c d e f

Codes (in ASCII): 97, 98, 97, 97, 97, 98, 98, 98, 99, 99, 100, 100, 100, 100, 99, 99

Codes (in MTF): 97, 98, 1, 0,

List

Index: 0 1 2 3 4 97 98 99 100 101 102 ... 255

Value: a b c d e f

11

Example: abaaabbbccdddddcc

List

Index: 0 1 2 3 4 97 98 99 100 101 102 ... 255

Value: a b c d e f

Codes (in ASCII): 97, 98, 97, 97, 97, 98, 98, 98, 99, 99, 100, 100, 100, 100, 99, 99

Codes (in MTF): 97, 98, 1, 0, 0, 1, 0, 0, 99, 0, 100, 0, 0, 0, 1, 0

12

Example: MTF decoding

13

Example: MTF decoding

14

Example: MTF decoding

15

Example: MTF decoding

16

Example: MTF decoding

17

BWT compressor vs ZIP

ZIP (i.e., LZW based)			BWT+RLE+MTF+AC		
File Name	Raw Size	PKZIP Size	PKZIP Bits/Byte	BWT Size	BWT Bits/Byte
blb	111,261	35,821	2.58	29,567	2.13
book1	768,771	315,999	3.29	275,831	2.87
book2	610,856	209,061	2.74	186,592	2.44
geo	102,400	68,917	5.38	62,120	4.85
news	377,109	146,010	3.10	134,174	2.85
obj1	21,504	10,311	3.84	10,857	4.04
obj2	246,814	81,846	2.65	81,948	2.66

18